

Activation functions

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Bias terms and decision boundaries

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Representing inputs as vectors

Without opening the old lesson, explain one key idea from Representing inputs as vectors. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Apply a declared class rule for Activation functions. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Activation functions. Proposed explanation: “ReLU returns a class label”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Trace a new chain independently

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Trace three activations

For scores -2, 0, 2, list identity and ReLU outputs. Explain why ReLU(2) is not 1.

My reasoning and evidence

6. W2 Apply a declared class rule

Use the lesson sigmoid values for -2, 0, 2. Predict classes under a ≥ 0.5 , then explain the zero-score case.

My reasoning and evidence

7. W3 Check the midpoint

Compare ReLU at inputs -2, 0, 2. Give the average of the two endpoint outputs and the output at the

midpoint; explain what this shows.

My reasoning and evidence

8. W4 Trace a new chain independently

Let $z = x - 2$, $a = \text{ReLU}(z)$, $y = 2a + 1$. For $x = 0, 2, 5$, record every z , a and y . Then replace ReLU with identity and give all three y values. Which inputs have the same ReLU output, and why?

My reasoning and evidence

9. W5 Limit a probability claim

A model returns a sigmoid value 0.9 on one case. Does the numerical range alone establish a calibrated 90 percent probability? Explain what needs checking.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled activation functions evidence sheet to Image-pattern Network Investigation. Show how you can trace identity relu and sigmoid on a numerical score. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Activation functions

See what ReLU removes before the next weighted step.

Toy network $z=x-3$; $h=\max(0, z)$; $y=3h+2$. Inputs $x=1, 3, 4$.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/4

ReLU hidden computation at $x=1$

Input 1 x weight 1 = 1

Add the products and bias -3: total = -2

ReLU: use $\max(0, \text{total}) \rightarrow \text{output } 0$

Activation functions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute preactivation

z values are -2, 0, 1.



2. Apply ReLU

h values are 0, 0, 1; the negative value is replaced by 0.



3. Compute output

y values are 2, 2, 5; x = 1 and x = 3 collapse to the same output.

CHANGE ONE THING

Replace ReLU with identity.

Explain the changed result using the matching lesson.

Which original inputs cannot be distinguished by this network's final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.